Part I

Getting Started & Configuration

Chapter 1

Getting Started with LinuxCNC

1.1 About LinuxCNC

1.1.1 The Software

- LinuxCNC (the Enhanced Machine Control) is a software system for computer control of machine tools such as milling machines and lathes, robots such as puma and scara and other computer controlled machines up to 9 axes.
- LinuxCNC is free software with open source code. Current versions of LinuxCNC are entirely licensed under the GNU General Public License and Lesser GNU General Public License (GPL and LGPL)
- LinuxCNC provides:
  - easy discovery and testing without installation with the LiveCD
  - easy installation from the Live CD
  - easy to use graphical configuration wizards to rapidly create a configuration specific to the machine
  - directly available as regular packages of recent releases of Debian (since Bookworm) and Ubuntu (since Kinetic Kudu)
  - a graphical user interface (actually several interfaces to choose from)
  - a graphical interface creation tool (Glade)
  - an interpreter for *G-code* (the RS-274 machine tool programming language)
  - a realtime motion planning system with look-ahead
  - operation of low-level machine electronics such as sensors and motor drives
  - an easy to use *breadboard* layer for quickly creating a unique configuration for your machine
  - a software PLC programmable with ladder diagrams
- It does not provide drawing (CAD - Computer Aided Design) or G-code generation from the drawing (CAM - Computer Automated Manufacturing) functions.
- It can simultaneously move up to 9 axes and supports a variety of interfaces.
- The control can operate true servos (analog or PWM) with the feedback loop closed by the LinuxCNC software at the computer, or open loop with step-servos or stepper motors.
- Motion control features include: cutter radius and length compensation, path deviation limited to a specified tolerance, lathe threading, synchronized axis motion, adaptive feedrate, operator feed override, and constant velocity control.

- Support for non-Cartesian motion systems is provided via custom kinematics modules. Available architectures include hexapods (Stewart platforms and similar concepts) and systems with rotary joints to provide motion such as PUMA or SCARA robots.
- LinuxCNC runs on Linux using real time extensions.

1.1.2 The Operating System

LinuxCNC is available as ready-to-use packages for the Ubuntu and Debian distributions.

1.1.3 Getting Help

1.1.3.1 IRC

IRC stands for Internet Relay Chat. It is a live connection to other LinuxCNC users. The LinuxCNC IRC channel is #linuxcnc on libera.chat.

The simplest way to get on the IRC is to use the embedded client on this page.

Some IRC etiquette

- Ask specific questions... Avoid questions like "Can someone help me?".
- If you're really new to all this, think a bit about your question before typing it. Make sure you give enough information so someone can solve your question.
- Have some patience when waiting for an answer, sometimes it takes a while to formulate an answer or everyone might be busy working or something.
- Set up your IRC account with your unique name so people will know who you are. If you use the java client, use the same name every time you log in. This helps people remember who you are and if you have been on before many will remember the past discussions which saves time on both ends.

Sharing Files

The most common way to share files on the IRC is to upload the file to one of the following or a similar service and paste the link:

- *For text*: https://pastebin.com/, http://pastie.org/, https://gist.github.com/
- *For pictures*: https://imagebin.org/, https://imgur.com/, https://bayimg.com/
- *For files*: https://filedropper.com/, https://filefactory.com/, https://1fichier.com/

1.1.3.2 Mailing List

An Internet Mailing List is a way to put questions out for everyone on that list to see and answer at their convenience. You get better exposure to your questions on a mailing list than on the IRC but answers take longer. In a nutshell you e-mail a message to the list and either get daily digests or individual replies back depending on how you set up your account.

You can subscribe to the emc-users mailing list at: https://lists.sourceforge.net/lists/listinfo/emc-users

1.1.3.3 Web Forum

A web forum can be found at https://forum.linuxcnc.org or by following the link at the top of the linuxcnc.org home page.

This is quite active but the demographic is more user-biased than the mailing list. If you want to be sure that your message is seen by the developers then the mailing list is to be preferred.

1.1.3.4 LinuxCNC Wiki

A Wiki site is a user maintained web site that anyone can add to or edit.

The user maintained LinuxCNC Wiki site contains a wealth of information and tips at: https://wiki.linuxcnc.or

1.1.3.5 Bug Reports

Report bugs to the LinuxCNC github bug tracker.

1.2 System Requirements

1.2.1 Minimum Requirements

The minimum system to run LinuxCNC and Debian / Ubuntu may vary depending on the exact usage. Stepper systems in general require faster threads to generate step pulses than servo systems. You can use the Live CD to test the software before committing to a permanent installation on a computer. Keep in mind that the Latency Test numbers are more important than the processor speed for software step generation. More information on the Latency Test is here. In addition LinuxCNC needs to be run on an operating system that uses a specially modified kernel, see Kernel and Version Requirements.

Additional information is on the LinuxCNC Wiki site: Hardware Requirements

LinuxCNC and Debian Linux should run reasonably well on a computer with the following minimum hardware specification. These numbers are not the absolute minimum but will give reasonable performance for most stepper systems.

- 700 MHz x86 processor (1.2 GHz x86 processor recommended) or Raspberry Pi 4 or better.
- LinuxCNC 2.8 or later from the Live CD expects a 64-bit capable system.
- 512 MB of RAM, 4 GB with GUI to avoid surprises
- No hard disk for Live CD, 8 GB or more for permanent installation
- Graphics card capable of at least 1024x768 resolution, which is not using the NVidia or ATI fglrx proprietary drivers. Modern onboard graphic chipsets seem to generally be OK.
- Internet connection (not strictly needed, but very useful for updates and for communicating with the LinuxCNC community)

Minimum hardware requirements change as Linux distributions evolve so check the Debian web site for details on the Live CD you're using. Older hardware may benefit from selecting an older version of the Live CD when available.

If you plan not to rely on the distribution of readily executable programs ("binaries") but aim at contributing to the source tree of LinuxCNC, then there is a good chance you want a second computer to perform the compilation. Even though LinuxCNC and your developments could likely be executed at the same time with respect to disk space, RAM and even CPU speed, a machine that is busy will have worse latencies, so you are unlikely to compile your source tree and produce chips at the same time.

1.2.2 Kernel and Version requirements

LinuxCNC requires a kernel modified for realtime use to control real machine hardware. It can, however run on a standard kernel in simulation mode for purposes such as checking G-code, testing config files and learning the system. To work with these kernel versions there are two versions of LinuxCNC distributed. The package names are ”linuxcnc” and ”linuxcnc-uspace”.

The realtime kernel options are preempt-rt, RTAI and Xenomai.

You can discover the kernel version of your system with the command:

```
uname -a
```

If you see (as above) `-rt-` in the kernel name then you are running the preempt-rt kernel and should install the ”uspace” version of LinuxCNC. You should also install uspace for ”sim” configs on non-realtime kernels

If you see `-rtai-` in the kernel name then you are running RTAI realtime. See below for the LinuxCNC version to install.

1.2.2.1 Preempt-RT with *linuxcnc-uspace* package

Preempt-RT is the newest of the realtime systems, and is also the version that is closest to a mainline kernel. Preempt-RT kernels are available as precompiled packages from the main repositories. The search term ”PREEMPT_RT” will find them, and one can be downloaded and installed just like any other package. Preempt-RT will generally have the best driver support and is the only option for systems using the Mesa ethernet-connected hardware driver cards. In general preempt-rt has the worst latency of the available systems, but there are exceptions.

1.2.2.2 RTAI with *linuxcnc* package

RTAI has been the mainstay of LinuxCNC distributions for many years. It will generally give the best realtime performance in terms of low latency, but might have poorer peripheral support and not as many screen resolutions. An RTAI kernel is available from the LinuxCNC package repository. If you installed from the Live/Install image then switching kernel and LinuxCNC flavour is described in [Installing-RTAI].

1.2.2.3 Xenomai with *linuxcnc-uspace* package

Xenomai is also supported, but you will have to find or build the kernel and compile LinuxCNC from source to utilise it.

1.2.2.4 RTAI with *linuxcnc-uspace* package

It is also possible to run LinuxCNC with RTAI in user-space mode. As with Xenomai you will need to compile from source to do this.

1.2.3 Problematic Hardware

1.2.3.1 Laptops

Laptops are not generally suited to real time software step generation. Again a Latency Test run for an extended time will give you the info you need to determine suitability.

1.2.3.2 Video Cards

If your installation pops up with 800 x 600 screen resolution then most likely Debian does not recognize your video card or monitor. This can sometimes be worked-around by installing drivers or creating / editing Xorg.conf files.

1.3 Getting LinuxCNC

This section describes the recommended way to download and make a fresh install of LinuxCNC. There are also Alternate Install Methods for the adventurous. If you have an existing install that you want to upgrade, go to the Updating LinuxCNC section instead.

Note
To operate machinery LinuxCNC requires a special kernel with real-time extensions. There are three possibilities here: preempt-rt, RTAI or Xenomai. In addition there are two versions of LinuxCNC which work with these kernels. See the table below for details. However for code testing and simulation it is possible to run the linuxcnc-uspace application on a stock kernel.

Fresh installs of LinuxCNC are most easily created using the Live/Install Image. This is a hybrid ISO filesystem image that can be written to a USB storage device or a DVD and used to boot a computer. At boot time you will be given a choice of booting the ”Live” system (to run LinuxCNC without making any permanent changes to your computer) or booting the Installer (to install LinuxCNC and its operating system onto your computer’s hard drive).

The outline of the process looks like this:

1. Download the Live/Install Image.
2. Write the image to a USB storage device or DVD.
3. Boot the Live system to test out LinuxCNC.
4. Boot the Installer to install LinuxCNC.

1.3.1 Download the image

This section describes some methods for downloading the Live/Install image.

1.3.1.1 Normal Download

Software for LinuxCNC to download is presented on the project’s Downloads page. Most users will aim for the disk image for Intel/AMD PCs, the URL will resemble https://www.linuxcnc.org/iso/-linuxcnc_2.9.4-amd64.hybrid.iso.

For the Raspberry Pi, multiple images are provided to address differences between the RPi4 and RPi5.

Note
Do not use the regular Raspbian distribution for LinuxCNC that may have shipped with your RPi starter kit - that will not have the real-time kernel and you cannot migrate from Raspbian to Debian’s kernel image.

1.3.1.2 Download using zsync

zsync is a download application that efficiently resumes interrupted downloads and efficiently transfers large files with small modifications (if you have an older local copy). Use zsync if you have trouble downloading the image using the Normal Download method.

zsync in Linux

1. Install zsync using Synaptic or, by running the following in a terminal

```
sudo apt-get install zsync
```

2. Then run this command to download the iso to your computer

```
zsync https://www.linuxcnc.org/iso/linuxcnc_2.9.4-amd64.hybrid.iso
```

zsync in Windows There is a Windows port of zsync. It works as a console application and can be downloaded from https://www.assembla.com/spaces/zsync-windows/documents .

1.3.1.3 Verify the image

(This step is unnecessary if you used zsync)

1. After downloading, verify the checksum of the image to ensure integrity.

```
md5sum linuxcnc-2.9.4-amd64.iso
```

or

```
sha256sum linuxcnc-2.9.4-amd64.iso
```

1. Then compare to these checksums

```
amd64 (PC)
md5sum: 1815aceaac0e7861747aa34d61846e79
sha256sum: 08b3f59233e47c91cf1c9a85c41df48542c97b134efefa7446d3060c9a3e644b
arm64 (Pi)
md5sum: 4547e8a72433efb033f0a5cf166a5cd2
sha256sum: ff3ba9b8dfb93baf1e2232746655f8521a606bc0fab91bffc04ba74cc3be6bf0
```

Verify md5sum on Windows or Mac Windows and Mac OS X do not come with an md5sum program, but there are alternatives. More information can be found at: How To MD5SUM

1.3.2 Write the image to a bootable device

The LinuxCNC Live/Install ISO Image is a hybrid ISO image which can be written directly to a USB storage device (flash drive) or a DVD and used to boot a computer. The image is too large to fit on a CD.

1.3.2.1 Raspberry Pi Image

The Raspbery Pi image is a completes SD card image and should be written to an SD card with the [Raspberry Pi Imager App](https://www.raspberrypi.com/software/).

1.3.2.2 AMD-64 (x86-64, PC) Image using GUI tools

Download and install [Balena Etcher](https://etcher.balena.io/#download-etcher) (Linux, Windows, Mac) and write the downloaded image to a USB drive.

If your image fails to boot then please also try Rufus. It looks more complicated but seems to be more compatible with various BIOSes.

1.3.2.3 Command line - Linux

1. Connect a USB storage device (for example a flash drive or thumb drive type device).
2. Determine the device file corresponding to the USB flash drive. This information can be found in the output of `dmesg` after connecting the device. `/proc/partitions` may also be helpful.
3. Use the `dd` command to write the image to your USB storage device. For example, if your storage device showed up as `/dev/sde`, then use this command:

```
dd if=linuxcnc_2.9.4-amd64.hybrid.iso of=/dev/sde
```

1.3.2.4 Command line - MacOS

1. Open a terminal and type

```
diskutil list
```

2. Insert the USB and note the name of the new disk that appears, eg /dev/disk5
3. unmount the USB. The number found above should be substitued in place of the N

```
diskutil unmountDisk /dev/diskN
```

4. Transfer the data with dd, as for Linux above. Note that the disk name has an added ”r” at the begining

```
sudo dd if=linuxcnc_2.9.4-amd64.hybrid.iso of=/dev/rdiskN bs=1m
```

5. Note that this may take a long time to complete and there will be no feedback during the process.

Writing the image to a DVD in Linux

1. Insert a blank DVD into your burner. A *CD/DVD Creator* or *Choose Disc Type* window will pop up. Close this, as we will not be using it.
2. Browse to the downloaded image in the file browser.
3. Right click on the ISO image file and choose Write to Disc.
4. Select the write speed. It is recommended that you write at the lowest possible speed.
5. Start the burning process.
6. If a *choose a file name for the disc image* window pops up, just pick OK.

Writing the image to a DVD in Windows

1. Download and install Infra Recorder, a free and open source image burning program: http://infrarecord

2. Insert a blank CD in the drive and select Do nothing or Cancel if an auto-run dialog pops up.
3. Open Infra Recorder, and select the *Actions* menu, then *Burn image*.

Writing the image to a DVD in Mac OSX

1. Download the .iso file
2. Right-click on the file in the Finder window and select ”Burn to disc” (The option to burn to disc will only appear if the machine has an optical drive fitted or connected)

1.3.3 Testing LinuxCNC

With the USB storage device plugged in or the DVD in the DVD drive, shut down the computer then turn the computer back on. This will boot the computer from the Live/Install Image and choose the Live boot option.

Note
If the system does not boot from the DVD or USB stick, it may be necessary to change the boot order in the PC BIOS.

Once the computer has booted up you can try out LinuxCNC without installing it. You can not create custom configurations or modify most system settings in a Live session, but you can (and should) run the latency test.

To try out LinuxCNC: from the Applications/CNC menu pick LinuxCNC. A dialog box will open from which you can choose one of many sample configurations. At this point it only really makes sense to pick a ”sim” configuration. Some of the sample configurations include onscreen 3D simulated machines, look for ”Vismach” to see these.

To see if your computer is suitable for software step pulse generation run the Latency Test as shown here.

At the time of writing the Live Image is only available with the preempt-rt kernel and a matching LinuxCNC. On some hardware this might not offer good enough latency. There is an experimental version available using the RTAI realtime kernel which will often give better latency.

1.3.4 Installing LinuxCNC

To install LinuxCNC from the Live CD select *Install (Graphical)* at bootup.

1.3.5 Updates to LinuxCNC

With the normal install the Update Manager will notify you of updates to LinuxCNC when you go on line and allow you to easily upgrade with no Linux knowledge needed. It is OK to upgrade everything except the operating system when asked to.



Warning
Do not upgrade the operating system if prompted to do so. You should accept OS *updates* however, especially security updates.

1.3.6 Install Problems

In rare cases you might have to reset the BIOS to default settings if during the Live CD install it cannot recognize the hard drive during the boot up.

1.3.7 Alternate Install Methods

The easiest, preferred way to install LinuxCNC is to use the Live/Install Image as described above. That method is as simple and reliable as we can make it, and is suitable for novice users and experienced users alike. However, this will typically replace any existing operating system. If you have files on the target PC that you want to keep, then use one of the methods described in this section.

In addition, for experienced users who are familiar with Debian system administration (finding install images, manipulating apt sources, changing kernel flavors, etc), new installs are supported on following platforms: ("amd64" means "64-bit", and is not specific to AMD processors, it will run on any 64-bit x86 system)

Distribution	Architecture	Kernel	Package name	Typical use
Debian Bookworm	amd64 & arm64	preempt-rt	linuxcnc-uspace	machine control & simulation
Debian Bookworm	amd64	RTAI	linuxcnc	machine control
Debian Bullseye	amd64	preempt-rt	linuxcnc-uspace	machine control & simulation
Debian Buster	amd64 & arm64	preempt-rt	linuxcnc-uspace	machine control & simulation
Debian Buster	amd64	RTAI	linuxcnc	machine control
Any	Any	Stock	linuxcnc-uspace	simulation ONLY

Note
LinuxCNC v2.9 is not supported on Debian 9 or older.

Preempt-RT kernels The Preempt-rt kernels are available for Debian from the regular debian.org archive. The package is called `linux-image-rt-*`. Simply install the package in the same way as any other package from the Synaptic Package manager or with apt-get at the command-line.

RTAI Kernels The RTAI kernels are available for download from the linuxcnc.org debian archive. The apt source is:

- Debian Bookworm: `deb http://linuxcnc.org bookworm base`
- Debian Bullseye: `deb http://linuxcnc.org bullseye base`
- Debian Buster: `deb http://linuxcnc.org buster base`

LinuxCNC and the RTAI kernel are now only available for 64-bit OSes but there are very few surviving systems that can not run a 64-bit OS.

1.3.7.1 Installing on Debian Bookworm (with Preempt-RT kernel)

1. Install Debian Bookworm (Debian 12), amd64 version. You can download the installer here: https://www.debian.org/distrib/

2. After burning the iso and booting up if you don't want Gnome desktop select *Advanced Options > Alternative desktop environments* and pick the one you like. Then select *Install* or *Graphical Install*.



Warning
Do not enter a root password, if you do sudo is disabled and you won't be able to complete the following steps.

3. Run the following in a terminal to bring the machine up to date with the latest packages.

```
sudo apt-get update
sudo apt-get dist-upgrade
```

Note
It is possible to download a version of LinuxCNC directly from Debian but this will install an old pre-release version, and is not recommended at this time.

1. Install the Preempt-RT kernel and modules

```
sudo apt-get install linux-image-rt-amd64
```

2. Re-boot, and select the Linux 6.1.0-10-rt-amd64 kernel. The exact kernel version might be different, look for the "-rt" suffix. This might be hidden in the "Advanced options for Debian Bookworm" sub-menu in Grub. When you log in, verify that `PREEMPT RT` is reported by the following command.

```
uname -v
```

3. Open Applications Menu > System > Synaptic Package Manager search for *linux-image* and right click on the original non-rt and select *Mark for Complete Removal*. Reboot. This is to force the system to boot from the RT kernel. If you prefer to retain both kernels then the other kernels need not be deleted, but grub boot configuration changes will be needed beyond the scope of this document.
4. Add the LinuxCNC Archive Signing Key to your apt keyring by downloading [the LinuxCNC installer script](https://www.linuxcnc.org/linuxcnc-install.sh) You will need to make the script executable to run it:

```
chmod +x linuxcnc-install.sh
```

Then you can run the installer:

```
sudo ./linuxcnc-install.sh
```

1.3.7.2 Installing on Debian Bookworm (with experimental RTAI kernel)

1. This kernel and LinuxCNC version can be installed on top of the Live DVD install, or alternatively on a fresh Install of Debian Bookworm 64-bit as described above.
2. You can add the LinuxCNC Archive signing key and repository information by downloading and running the installer script as decribed above. If an RTAI kernel is detected it will stop before installing any packages.

3. Update the package list from linuxcnc.org

```
sudo apt-get update
```

4. Remove the existing uspace version of LinuxCNC and install the new realtime kernel, RTAI and the rtai version of linuxcnc.

```
sudo apt-get purge linuxcnc-uspace
sudo apt-get purge linuxcnc-doc*
sudo apt-get install linuxcnc
```

Reboot the machine, ensuring that the system boots from the new 5.4.258-rtai kernel.

1.3.7.3 Installing on Raspbian 12

Don't do that. The latencies are too bad with the default kernel and the PREEMPT_RT (the RT is important) kernel of Debian does not boot on the Pi (as of 1/2024). Please refer to the images provided online. You can create them yourself following the scripts provided online.

1.4 Running LinuxCNC

1.4.1 Invoking LinuxCNC

After installation, LinuxCNC starts just like any other Linux program: run it from the terminal by issuing the command *linuxcnc*, or select it in the *Applications -> CNC* menu.

1.4.2 Configuration Launcher

When starting LinuxCNC (from the CNC menu or from the command line without specifying an INI file) the Configuration Selector dialog starts.

The Configuration Selector dialog allows the user to pick one of their existing configurations (My Configurations) or select a new one (from the Sample Configurations) to be copied to their home directory. Copied configurations will appear under My Configurations on the next invocation of the Configuration Selector.

The Configuration Selector offers a selection of configurations organized:

- *My Configurations* - User configurations located in linuxcnc/configs in your home directory.
- *Sample Configurations* - Sample configurations, when selected, are copied to linuxcnc/configs. Once a sample configuration was copied to your local directory, the launcher will offer it as *My Configurations*. The names under which these local configurations are presented correspond to the names of the directories within the configs/ directory:
  - *sim* - Configurations that include simulated hardware. These can be used for testing or learning how LinuxCNC works.
  - *by_interface* - Configurations organized by GUI.
  - *by_machine* - Configurations organized by machine.
  - *apps* - Applications that do not require starting linuxcnc but may be useful for testing or trying applications like PyVCP or GladeVCP.
  - *attic* - Obsolete or historical configurations.

The sim configurations are often the most useful starting point for new users and are organized around supported GUIs:

- *axis* - Keyboard and Mouse GUI
- *craftsman* - Touch Screen GUI (no longer maintained ???)
- *gmoccapy* - Touch Screen GUI
- *gscreen* - Touch Screen GUI
- *pyvcp_demo* - Python Virtual Control Panel
- *qtaxis* - Touch Screen GUI, axis lookalike
- *qtdragon* - Touch Screen GUI
- *qtdragon_hd* - Touch Screen GUI, high definition
- *qtplasmac* - Touch Screen GUI, for plasma tables
- *qttouchy* - Touch Screen GUI
- *tklinuxcnc* - Keyboard and Mouse GUI (no longer maintained)
- *touchy* - Touch Screen GUI
- *woodpecker* - Touch Screen GUI A GUI configuration directory may contain subdirectories with configurations that illustrate special situations or the embedding of other applications.

The *by_interface* configurations are organized around common, supported interfaces like:

- general mechatronics
- mesa
- parport
- pico
- pluto
- servotogo
- vigilant
- vitalsystems

Related hardware may be required to use these configurations as starting points for a system.

The *by_machine* configurations are organized around complete, known systems like:

- boss
- cooltool
- scortbot erIII
- sherline
- smithy
- tormach

A complete system may be required to use these configurations.

The *apps items* are typically either:

1. utilities that don't require starting linuxcnc
2. demonstrations of applications that can be used with linuxcnc
   - info - creates a file with system information that may be useful for problem diagnosis.
   - gladevcp - Example GladeVCP applications.
   - halrun - Starts halrun in an terminal.
   - latency - Applications to investigate latency
     - latency-histogram-1 - histogram for single servo thread
     - latency-histogram - histogram
     - latency-test - standard test
     - latency-plot - stripchart
   - parport - Applications to test parport.
   - pyvcp - Example pyvcp applications.
   - xhc-hb04 - Applications to test an xhc-hb04 USB wireless MPG

Note

Under the Apps directory, only applications that are usefully modified by the user are offered for copying to the user's directory.

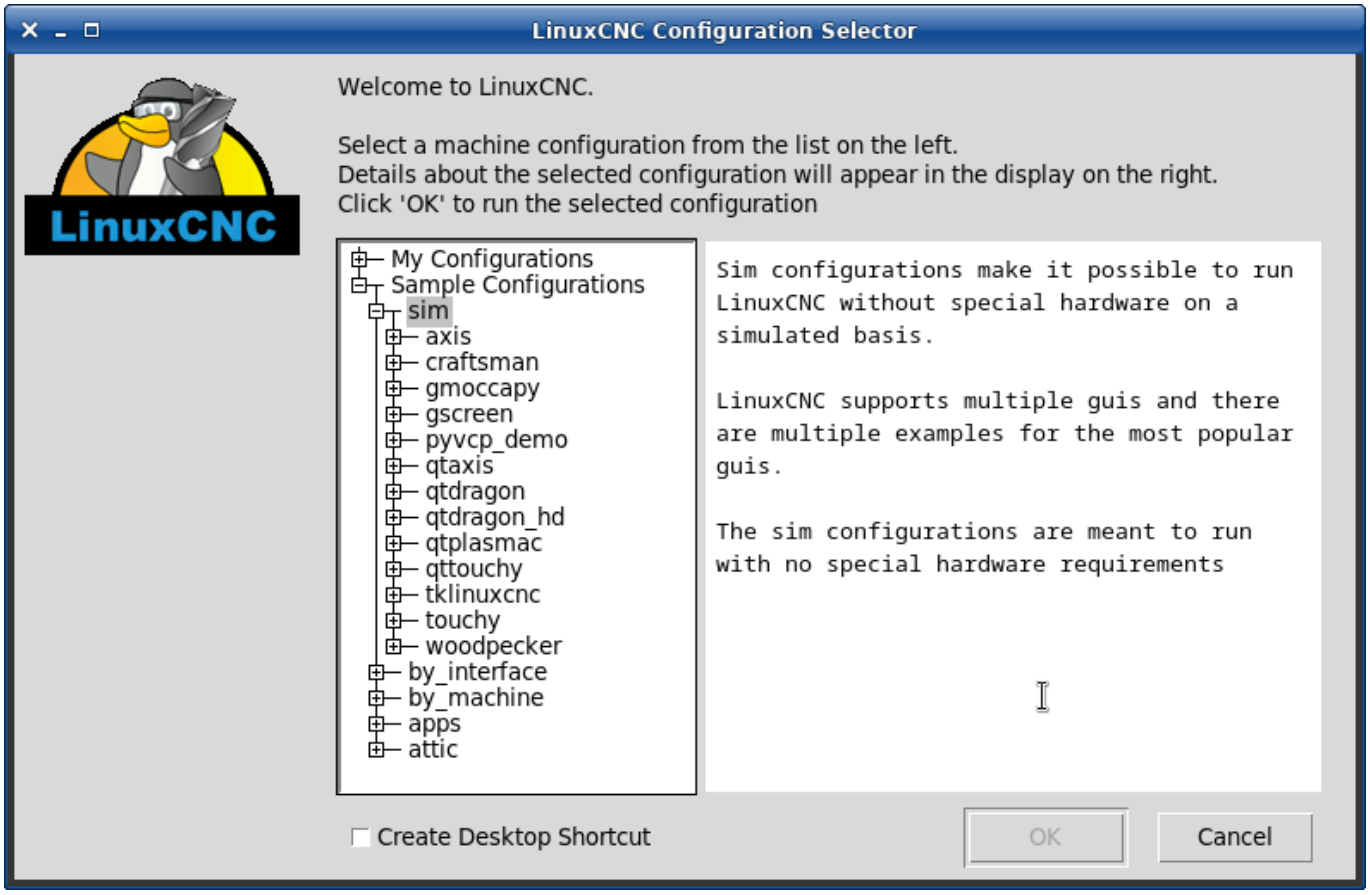


Figure 1.1: LinuxCNC Configuration Selector

Click any of the listed configurations to display specific information about it. Double-click a configuration or click OK to start the configuration.

Select *Create Desktop Shortcut* and then click *OK* to add an icon on the Ubuntu desktop to directly launch this configuration without showing the Configuration Selector screen.

When you select a configuration from the Sample Configurations section, it will automatically place a copy of that config in the ~/linuxcnc/configs directory.

1.4.3 Next steps in configuration

After finding the sample configuration that uses the same interface hardware as your machine (or a simulator configuration), and saving a copy to your home directory, you can customize it according to the details of your machine. Refer to the Integrator Manual for topics on configuration.

1.4.4 Simulator Configurations

All configurations listed under Sample Configurations/sim are intended to run on any computer. No specific hardware is required and real-time support is not needed.

These configurations are useful for studying individual capabilities or options. The sim configurations are organized according to the graphical user interface used in the demonstration. The directory for axis contains the most choices and subdirectories because it is the most tested GUI. The capabilities demonstrated with any specific GUI may be available in other GUIs as well.

1.4.5 Configuration Resources

The Configuration Selector copies all files needed for a configuration to a new subdirectory of ~/linuxcnc/configs (equivalently: /home/username/linuxcnc/configs). Each created directory will include at least one INI file (iniflename.ini) that is used to describe a specific configuration.

File resources within the copied directory will typically include one or more INI file (filename.ini) for related configurations and a tool table file (toolfilename.tbl). Additionally, resources may include HAL files (filename.hal, filename.tcl), a README file for describing the directory, and configuration specific information in a text file named after a specific configuration (inifilename.txt). That latter two files are displayed when using the Configuration Selector.

The supplied sample configurations may specify the parameter HALFILE (filename.hal) in the configuration INI file that are not present in the copied directory because they are found in the system HAL file library. These files can be copied to the user configuration directory and altered as required by the user for modification or test. Since the user configuration directory is searched first when finding HAL files, local modifications will then prevail.

The Configuration selector makes a symbolic link in the user configuration directory (named hallib) that points to the system HAL file library. This link simplifies copying a library file. For example, to copy the library core_sim.hal file in order to make local modifications:

```
cd ~/linuxcnc/configs/name_of_configuration
cp hallib/core_sim.hal core_sim.hal
```

1.5 Updating LinuxCNC

Updating LinuxCNC to a new minor release (ie to a new version in the same stable series, for example from 2.9.1 to 2.9.2) is an automatic process if your PC is connected to the internet. You will see an update prompt after a minor release along with other software updates. If you don’t have an internet connection to your PC see Updating without Network.

1.5.1 Upgrade to the new version

This section describes how to upgrade LinuxCNC from version 2.8.x to a 2.9.y version. It assumes that you have an existing 2.8 install that you want to update.

To upgrade LinuxCNC from a version older than 2.8, you have to first upgrade your old install to 2.8, then follow these instructions to upgrade to the new version.

If you do not have an old version of LinuxCNC to upgrade, then you're best off making a fresh install of the new version as described in the section Getting LinuxCNC.

Furthermore, if you are running Ubuntu Precise or Debian Wheezy it is well worth considering making a backup of the "linuxcnc" directory on removable media and performing a clean install of a newer OS and LinuxCNC version as these releases were EOL in 2017 and 2018 respectively. If you are running on Ubuntu Lucid then you will have to do this, as Lucid is no longer supported by LinuxCNC (it was EOL in 2013).

To upgrade major versions like 2.8 to 2.9 when you have a network connection at the machine you need to disable the old linuxcnc.org apt sources in the file /etc/apt/sources.list and add a new linuxcnc.org apt source for 2.9, then upgrade LinuxCNC.

The details will depend on which platform you're running on. Open a terminal then type `lsb_release -ic` to find this information out:

```
lsb_release -ic
Distributor ID: Debian
Codename:       Buster
```

You should be running on Debian Buster, Bullseye or Bookworm or Ubuntu 20.04 "Focal Fossa" or newer. LinuxCNC 2.9.y will not run on older distributions than these.

You will also need to check which realtime kernel is being used:

```
uname -r
6.1.0-10-rt-amd64
```

If you see (as above) `-rt-` in the kernel name then you are running the preempt-rt kernel and should install the "uspace" version of LinuxCNC. You should also install uspace for "sim" configs on non-realtime kernels

If you see `-rtai-` in the kernel name then you are running RTAI realtime. See below for the LinuxCNC version to install. RTAI packages are available for Bookworm and Buster but not currently for Bullseye.

1.5.1.1 Apt Sources Configuration

- Open the `Software Sources` window. The process for doing this differs slightly on the three supported platforms:
  - Debian:
    - Click on `Applications Menu`, then `System`, then `Synaptic Package Manager`.
    - In Synaptic, click on the `Settings` menu, then click `Repositories` to open the `Software Sources` window.
  - Ubuntu Precise:
    - Click on the `Dash Home` icon in the top left.
    - In the `Search` field, type "software", then click on the `Ubuntu Software Center` icon.
    - In the Ubuntu Software Center window, click on the `Edit` menu, then click on `Software Sources...` to open the `Software Sources` window.
  - Ubuntu Lucid:

  * Click the `System` menu, then `Administration`, then `Synaptic Package Manager`.
  * In Synaptic, click on the `Settings` menu, then click on `Repositories` to open the `Software Sources` window.

- In the `Software Sources` window, select the `Other Software` tab.
- Delete or un-check all the old linuxcnc.org entries (leave all non-linuxcnc.org lines as they are).
- Click the `Add` button and add a new apt line. The line will be slightly different on the different platforms:

Table 1.2: Tabular overview on variants of the Operating System and the corresponding configuration of the repository. The configuration can be performed in the GUI of the package manager or in the file /etc/apt/sources.list.

OS / Realtime Version	**Repository**
Debian Buster - preempt	`deb https://linuxcnc.org buster base 2.9-uspace`
Debian Buster - RTAI	`deb https://linuxcnc.org buster base 2.9-rt`
Debian Bullseye - preempt	`deb https://linuxcnc.org bullseye base 2.9-uspace`
Debian Bookworm - preempt	`deb https://linuxcnc.org bookworm base 2.9-uspace`
Debian Bookworm - RTAI	`deb https://linuxcnc.org bookworm base 2.9-rt`

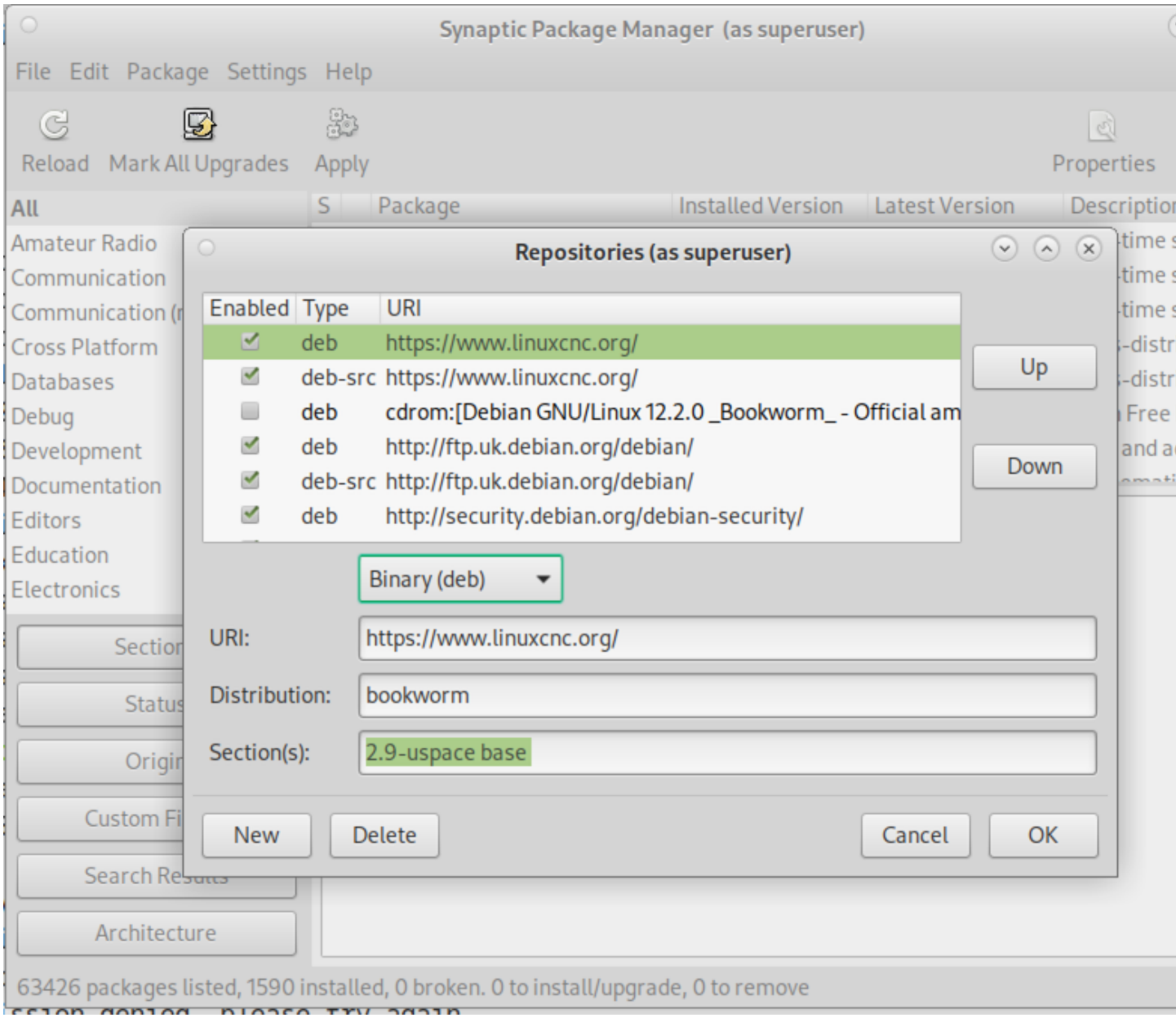


Figure 1.2: Figure with a screenshot of the repository configuration of the synaptic package manager.

- Click `Add Source`, then `Close` in the Software Sources window. If it pops up a window informing you that the information about available software is out-of-date, click the `Reload` button.

1.5.1.2 Upgrading to the new version

Now your computer knows where to get the new version of the software, next we need to install it.

The process again differs depending on your platform.

Debian uses the Synaptic Package Manager.

- Open Synaptic using the instructions in Setting apt sources above.
- Click the `Reload` button.

- Use the Search function to search for `linuxcnc`.
- The package is called ”linuxcnc” for RTAI kernels and ”linuxcnc-uspace” for preempt-rt.
- Click the check box to mark the new linuxcnc and linuxcnc-doc-* packages for upgrade. The package manager may select a number of additional packages to be installed, to satisfy dependencies that the new linuxcnc package has.
- Click the `Apply` button, and let your computer install the new package. The old linuxcnc package will be automatically upgraded to the new one.

1.5.1.3 Ubuntu

- Click on the `Dash Home` icon in the top left.
- In the `Search` field, type ”update”, then click on the `Update Manager` icon.
- Click the `Check` button to fetch the list of packages available.
- Click the `Install Updates` button to install the new versions of all packages.

1.5.2 Updating without Network

To update without a network connection you need to download the .deb then install it with dpkg. The .debs can be found in https://linuxcnc.org/dists/ .

You have to drill down from the above link to find the correct deb for your installation. Open a terminal and type in *lsb_release -ic* to find the release name of your OS.

```
> lsb_release -ic
Distributor ID: Debian
Codename:       bullseye
```

Pick the OS from the list then pick the major version you want like 2.9-rt for RTAI or 2.9-rtpreempt or 2.9-uspace for preempt-rt.

Next pick the type of computer you have: binary-amd64 for any 64-bit x86, binary-i386 for 32 bit, binary-armhf (32bit) or binary-arm64 (64bit) for Raspberry Pi.

Next pick the version you want from the bottom of the list like *linuxcnc-uspace_2.8.0_amd64.deb* (choose the latest by date). Download the deb and copy it to your home directory. You can rename the file to something a bit shorter with the file manager like *linuxcnc_2.9.2.deb* then open a terminal and install it with the package manager with this command:

```
sudo dpkg -i linuxcnc_2.9.2.deb
```

1.5.3 Updating Configuration Files for 2.9

1.5.3.1 Stricter handling of pluggable interpreters

If you just run regular G-code and you don’t know what a pluggable interpreter is, then this section does not affect you.

A seldom-used feature of LinuxCNC is support for pluggable interpreters, controlled by the undocumented `[TASK]INTERPRETER` INI setting.

Versions of LinuxCNC before 2.9.0 used to handle an incorrect `[TASK]INTERPRETER` setting by automatically falling back to using the default G-code interpreter.

Since 2.9.0, an incorrect `[TASK]INTERPRETER` value will cause LinuxCNC to refuse to start up. Fix this condition by deleting the `[TASK]INTERPRETER` setting from your INI file, so that LinuxCNC will use the default G-code interpreter.

1.5.3.2 Canterp

If you just run regular G-code and you don't use the `canterp` pluggable interpreter, then this section does not affect you.

In the extremely unlikely event that you are using `canterp`, know that the module has moved from `/usr/lib/libcanterp.so` to `/usr/lib/linuxcnc/canterp.so`, and the `[TASK]INTERPRETER` setting correspondingly needs to change from `libcanterp.so` to `canterp.so`.

1.5.4 Updating Configuration Files (for 2.9.y)

No changes should be necessary to configuration files when moving from 2.8.x to 2.9.y.

1.5.4.1 Spindle limits in the INI

It is now possible to add settings to the [SPINDLE] section of the INI file

MAX_FORWARD_VELOCITY = 20000 The maximum spindle speed (in rpm)

MIN_FORWARD_VELOCITY = 3000 The minimum spindle speed (in rpm)

MAX_REVERSE_VELOCITY = 20000 This setting will default to MAX_FORWARD_VELOCITY if omitted.

MIN_REVERSE_VELOCITY = 3000` This setting is equivalent to MIN_FORWARD_VELOCITY but for reverse spindle rotation. It will default to the MIN_FORWARD_VELOCITY if omitted.

INCREMENT = 200 Sets the step size for spindle speed increment / decrement commands. This can have a different value for each spindle. This setting is effective with AXIS and Touchy but note that some control screens may handle things differently.

HOME_SEARCH_VELOCITY = 100 - Accepted but currently does nothing

HOME_SEQUENCE = 0 - Accepted but currently does nothing

1.5.5 New HAL components

1.5.5.1 Non-Realtime

mdro mqtt-publisher pi500_vfd pmx485-test qtplasmac-cfg2prefs qtplasmac-materials qtplasmac-plasmac2c qtplasmac-setup sim-torch svd-ps_vfd

1.5.5.2 Realtime

anglejog div2 enum filter_kalman flipflop hal_parport homecomp limit_axis mesa_uart millturn scaled_s32_su tof ton

1.5.6 New Drivers

A framework for controlling ModBus devices using the serial ports on many Mesa cards has been introduced. http://linuxcnc.org/docs/2.9/html/drivers/mesa_modbus.html

A new GPIO driver for any GPIO which is supported by the gpiod library is now included: http://linuxcnc.org/-docs/2.9/html/drivers/hal_gpio.html

1.6 Linux FAQ

These are some basic Linux commands and techniques for new to Linux users. More complete information can be found on the web or by using the man pages.

1.6.1 Automatic Login

1.6.1.1 Debian

Debian Stretch uses the Xfce desktop environment by default, with the lightDM display manager lightDM. To get automatic login with Stretch:

- In a terminal, use the command:

```
$ /usr/sbin/lightdm --show-config
```

- Make a note of the absolute path to the configuration file lightdm.conf.
- Edit that file with a pure text editor (gedit, nano, etc), as root.
- Find and uncomment the lines:

```
#autologin-user=
#autologin-user-timeout=0
```

- Set autologin-user=your_user_name
- Save and reboot.

1.6.1.2 Ubuntu

When you install LinuxCNC with the Ubuntu LiveCD the default is to have to log in each time you turn the computer on. To enable automatic login go to *System* > *Administration* > *Login Window*. If it is a fresh install the Login Window might take a second or three to pop up. You will have to have your password that you used for the install to gain access to the Login Window Preferences window. In the Security tab check off Enable Automatic Login and pick a user name from the list (that would be you).

1.6.2 Automatic Startup

To have LinuxCNC start automatically with your config after turning on the computer go to *System* > *Preferences* > *Sessions* > *Startup Applications*, click Add. Browse to your config and select the .ini file. When the file picker dialog closes, add linuxcnc and a space in front of the path to your .ini file.

Example:

```
linuxcnc /home/mill/linuxcnc/config/mill/mill.ini
```

The documentation refers to your respective .ini file as INI-file.

1.6.3 Terminal

Many things need to be done from the terminal like checking the kernel message buffer with *dmesg*. Ubuntu and Linux Mint have a keyboard shortcut Ctrl + Alt + t. Debian Stretch does not have any keyboard shortcuts defined. It can be easily created with the *Configuration Manager*. Most modern file managers support the right key to open a terminal just make sure your right clicking on a blank area or a directory not a file name. Most OS's have the terminal as a menu item, usually in Accessories.

1.6.4 Man Pages

A man page (short for manual page) is a form of software documentation usually found on a Unix or Unix-like operating system like Linux.

To view a man page open up a terminal to find out something about the find command in the terminal window type:

```
man find
```

Use the Page Up and Page Down keys to view the man page and the Q key to quit viewing.

Note
Viewing the man page from the terminal may not get the expected man page. For example if you type in man abs you will get the C abs not the LinuxCNC abs. It is best to view the LinuxCNC man pages in the HTML documents.

1.6.5 List Modules

Sometimes when troubleshooting you need to get a list of modules that are loaded. In a terminal window type:

```
lsmod
```

If you want to send the output from lsmod to a text file in a terminal window type:

```
lsmod > mymod.txt
```

The resulting text file will be located in the home directory if you did not change directories when you opened up the terminal window and it will be named mymod.txt or what ever you named it.

1.6.6 Editing a Root File

When you open the file browser and you see the owner of the file is root you must do extra steps to edit that file. Editing some root files can have bad results. Be careful when editing root files. Generally, you can open and view most root files, but they will open in *read only* mode.

1.6.6.1 The Command Line Way

Open a terminal and type

```
sudo gedit
```

Open the file with File > Open > Edit

1.6.6.2 The GUI Way

1. Right click on the desktop and select Create Launcher
2. Type a name in like sudo edit
3. Type *gksudo "gnome-open %u"* as the command and save the launcher to your desktop
4. Drag a file onto your launcher to open and edit

1.6.6.3 Root Access

In Ubuntu you can become root by typing in "sudo -i" in a terminal window then typing in your password. Be careful, because you can really foul things up as root if you don't know what you're doing.

1.6.7 Terminal Commands

1.6.7.1 Working Directory

To find out the path to the present working directory in the terminal window type:

```
pwd
```

1.6.7.2 Changing Directories

To change the working directory to the one one level up, i.e., the parent directory, in the terminal window type:

```
cd ..
```

To move up two levels in the terminal window type:

```
cd ../..
```

To move directly to your home directory, in the terrminal window use the cd command with no arguments:

```
cd
```

To move down to the linuxcnc/configs subdirectory in the terminal window type:

```
cd linuxcnc/configs
```

1.6.7.3 Listing files in a directory

To view a list of all the files and subdirectories in the terminal window type:

```
dir
```

or

```
ls
```

1.6.7.4 Finding a File

The find command can be a bit confusing to a new Linux user. The basic syntax is:

```
find starting-directory parameters actions
```

For example to find all the .ini files in your linuxcnc directory you first need to use the pwd command to find out the directory.

Open a new terminal window and type:

```
pwd
```

And pwd might return the following result:

```
/home/joe
```

With this information put the command together like this:

```
find /home/joe/linuxcnc -name \*.ini -print
```

The -name is the name of the file your looking for and the -print tells it to print out the result to the terminal window. The *.ini tells find to return all files that have the .ini extension. The backslash is needed to escape the shell meta-characters. See the find man page for more information on find.

1.6.7.5 Searching for Text

```
grep -irl 'text to search for' *
```

This will find all the files that contain the *text to search for* in the current directory and all the subdirectories below it, while ignoring the case. The -i is for ignore case and the -r is for recursive (include all subdirectories in the search). The -l option will return a list of the file names, if you leave the -l off you will also get the text where each occurrence of the "text to search for" is found. The * is a wild card for search all files. See the grep man page for more information.

1.6.7.6 Diagnostic Messages

To view the diagnostic messages use "dmesg" from the command window. To save the diagnostic messages to a file use the redirection operator >, like this:

```
dmesg > bootmsg.txt
```

The contents of this file can be copied and pasted on line to share with people trying to help you diagnose your problem.

To clear the message buffer type this:

```
sudo dmesg -c
```

This can be helpful to do just before launching LinuxCNC, so that there will only be a record of information related to the current launch of LinuxCNC.

To find the built in parallel port address use grep to filter the information out of dmesg.

After boot up open a terminal and type:

```
dmesg|grep parport
```

1.6.8 Convenience Items

1.6.8.1 Terminal Launcher

If you want to add a terminal launcher to the panel bar on top of the screen you typically can right click on the panel at the top of the screen and select ”Add to Panel”. Select Custom Application Launcher and Add. Give it a name and put gnome-terminal in the command box.

1.6.9 Hardware Problems

1.6.9.1 Hardware Info

To find out what hardware is connected to your motherboard in a terminal window type:

```
lspci -v
```

1.6.9.2 Monitor Resolution

During installation Ubuntu attempts to detect the monitor settings. If this fails you are left with a generic monitor with a maximum resolution of 800x600.

Instructions for fixing this are located here:

https://help.ubuntu.com/community/FixVideoResolutionHowto

1.6.10 Paths

Relative Paths Relative paths are based on the startup directory which is the directory containing the INI-file. Using relative paths can facilitate relocation of configurations but requires a good understanding of linux path specifiers.

```
./f0        is the same as f0, e.g., a file named f0 in the startup directory
../f1       refers to a file f1 in the parent directory
../../f2    refers to a file f2 in the parent of the parent directory
../../../f3 etc.
```